



**Politecnico
di Torino**

Department of Electronics and Telecommunications
Bachelor degree in Electronic Engineering

11APMNX - Dispositivi Elettronici

Lab assignment 1: Homogeneous Semiconductor

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Intro duction

The first laboratory assignment is composed of different exercises to be solved autonomously using the PADRE tool on nanoHUB. In particular, it is focused on the analysis of a homogeneous silicon sample under different working conditions. The starting point is the equilibrium condition, then a bias voltage must be applied in order to analyse the device behaviour out of equilibrium. Lastly a temperature variation is added to highlight its effect on some physical quantities. After this preliminary study, you are required to add a light source to stimulate the system.

In order to get an evaluation, you **must** write a brief report in which you insert all the main outcomes obtained from the simulations together with an explanation and a comparison with the theoretical results. At the end of the report, you also have to insert all the PADRE input files necessary to perform the simulations. In the report follow the same structure of the assignment: highlight the answers by writing close to them the number of the corresponding question.

The report must be submitted in ONLY 1 PDF on the “Portale della Didattica > Elaborati”.

The report of this first assignment **must** be submitted within the **20th January 2026**.

For each exercise, specifications will be calculated considering some of your personal information: name, surname e student ID number. From now on

- **#n**: is the number of characters of your name;
- **#s**: is the number of characters of your surname.

For example, considering “Mario Bianchi”, #n = 5 and #s = 7.

IMPORTANT: Please write at the beginning of the report the specifications that you obtain using your personal information. Report both #n and #s and also the derived quantities like doping values.

IMPORTANT: Always specify in your answers the unit of measurement of all the quantities, in both graphs and formula results. The measurement units should be written in print (i.e., block letters), NOT in italic; a space should be present between the number (physical quantity value) and the measurement unit.

The report can be either in English or in Italian according to your preference.

Report also the PADRE input files you develop to do the assignment. They have to be submitted as screenshots at the end of the assignment

Important Remark: To assign you the data of the exercises, we will make use of the mod operation. It is the remainder after dividing one number by another. Example: $100 \bmod 9$ equals 1. Because $100/9 = 11$ with a remainder of 1.

EXERCISE A: homogeneous silicon not illuminated

Consider an uniformly doped silicon sample at 300 K. The doping type and amount are calculated using the equations below:

- **TYPE:** if $(\#n + \#s) \bmod 2 = 0$ then doping is n-type, otherwise is p-type¹;
- **AMOUNT:** doping concentration is $N = a \cdot 10^{16} \text{ cm}^{-3}$, with $a = \#s \bmod 9 + 1$.

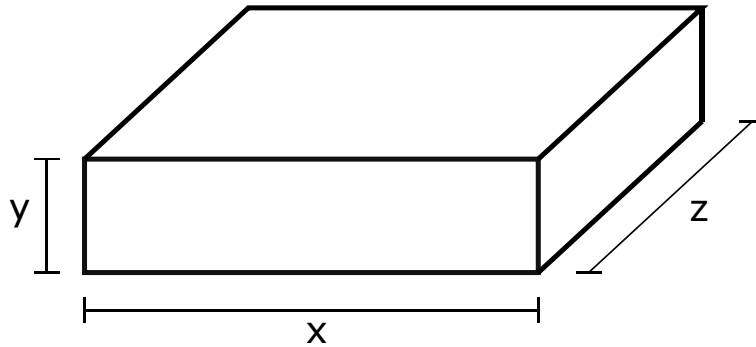


Figure 1: Dimensions of the silicon sample.

Considering Figure 1, the dimensions of the sample are:

- **x:** is $(\#n + \#s)$ times the sum of the last 4 digits of your student ID number, considered as μm ;
- **y:** is $\#s$ times the sum of the first 2 digits of your student ID number, considered as μm ;
- **z:** is $1 \mu\text{m}$ (PADRE default value).

For example, considering “Mario Bianchi” and “s123456”, $x = 216 \mu\text{m}$ and $y = 21 \mu\text{m}$.

IMPORTANT: Remember to report the specifications you use in your report!

Questions

Q A.1: plot the energy band diagram obtained from PADRE(E_f , E_{fi} , E_c , E_v); compare the energy-distances with the theoretical results($E_c - E_f$, $E_f - E_v$, $E_f - E_{fi}$).

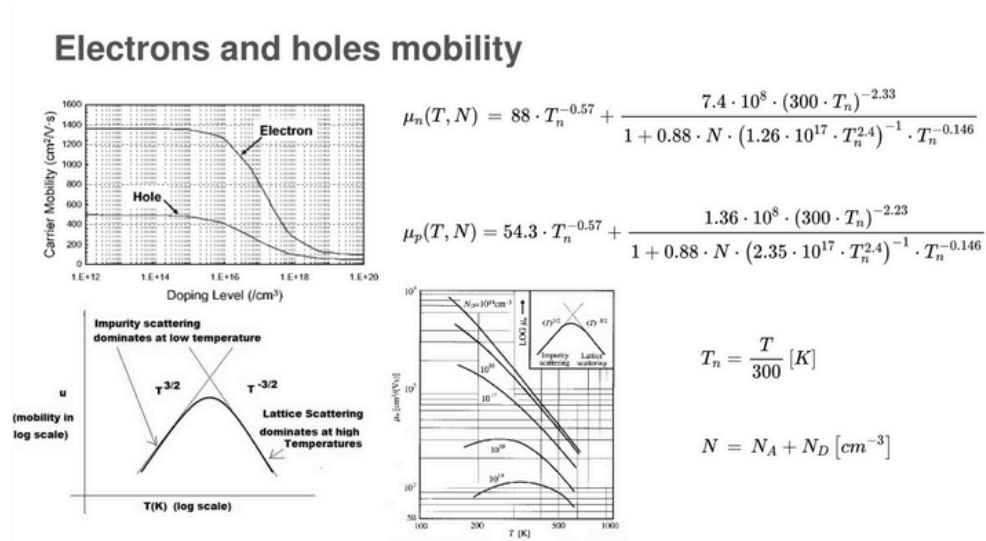
Q A.2: plot the electrons and holes concentrations obtained with PADRE and compare them with the theoretical values. Consider to compare them also graphically.

¹The function *mod* is the **modulo operation**, which returns the remainder of a division, where the number before the operator is divided by the one after the operator.

Q A.3: apply a positive bias $V_a = \#n/\#s$ V (rounded to the first decimal digit), then:

1. plot the energy band diagram obtained from PADRE(E_f , E_{fi} , E_c , E_v); compare the energy-distances with the theoretical results(E_c-E_f , E_f-E_v , E_f-E_{fi}).
2. plot the electric field obtained from PADRE and compare the obtained value with the theoretical one, commenting on it;
3. plot the IV-characteristic-curve and numerically evaluate the resistance of the sample through a liner fit. Compare it with the teoretical results.

Q A.4: repeat Q A.1 and Q A.2 bringing the temperature to 600 K. Which quantities change? How? (Use the following relation to evaluate the mobility for $T \neq 300K$)



HINT: use ALWAYS this relation to evaluate n_i

$$n_i^2 = N_c N_v e^{-E_g/k_B T}$$

EXERCISE B: homogeneous silicon illuminated

Consider now the same sample of silicon (at 300 K) of the previous exercise illuminated uniformly with a light source incident on the yz plane at $x = 0$ along the x-axis, with an optical generation rate equal to $G_{\text{opt}} = \#s \cdot 10^a \text{cm}^{-3}\text{s}^{-1}$, where $a = 20 + (\#n \cdot \#s) \bmod 3$.

Questions

Q B.1: considering the effect of light only (no bias):

1. calculate the diffusion length and compare it with the length of the sample, which approximation holds?(long/medium/short sample approximation)
2. plot the minority carrier concentration obtained with PADRE and compare the obtained values with the theoretical ones, commenting on it. Also, extract from the simulation results the light-induced excess carrier densities.
3. (**facultative**) calculate the diffusion length and compare it with the one extracted from the graph obtained from PADRE.(working with the tangent of the exponential in 0)

Q B.2: try now to fix the sample length to $\#s \mu\text{m}$ (x-dimension) and evaluate the same quantities of Q B.1 (consider only minority carriers), how do they change?

Are the results coherent with your expectations? Explain why.

Q B.3: now consider the effect of both light and bias, applying the following voltages $V_a = [-1; -0.5; 0.5; 1] \text{ V}$, then

1. plot the hole and electron concentrations obtained with PADRE and compare the obtained values with the theoretical ones, commenting on it. Compare the results obtained for the different applied voltages, how does V_a influence the carriers concentrations?
2. plot the total current obtained from PADRE and compare it to the theoretical value.